

EM-seq Frequently Asked Questions

Q1: What factors influence the mapping rate? Typically, what is considered a good mapping rate? Why is the mapping rate in an EM-seq project generally lower than that of a transcriptome analysis using the same sample?

A1: First, the mapping rate in the final report refers to the **unique mapping rate** (paired-end alignment), which excludes reads with multiple alignments and single-end aligned reads. By default, the **Bismark** software uses relatively stringent alignment parameters, ensuring high accuracy in the results.

Second, enzymatic treatment of DNA requires read conversion during alignment, which reduces sequence complexity (lower entropy) and can negatively impact the mapping rate.

Additionally, other factors can influence the mapping rate, such as:

- The completeness of the reference genome assembly
- The abundance of repetitive sequences in the genome
- The evolutionary distance between the reference genome and the sample
- Sample heterozygosity
- Different sample processing methods
- Sample contamination

For methylation projects, the **coverage** and **depth** of the data used for analysis are more critical. If sufficient data is obtained and coverage differences among samples are minimal, reliable detection of mC sites and differential analysis among samples can be ensured.

Regarding the difference between EM-seq and transcriptome mapping rates: EM-seq involves DNA-level alignment, while transcriptome sequencing involves RNA-level alignment. The alignment software used for RNA-seq is designed to account for splicing events and applies more lenient criteria, resulting in a higher mapping rate compared to methylation data.

Q2: How is methylation density calculated?

A2: There are multiple methods for defining and calculating methylation density. Here, we use the **relative density** approach: the number of methylated cytosines (mC)

in a given region is divided by the total number of cytosines (C) covered by $\geq 5X$ reads (the baseline for methylation calling) in that same region.

Q3: How is methylation level calculated?

A3: There are multiple methods for defining and calculating methylation level. Here, we adopt the **Weighted Methylation Level** approach from Schultz et al., 2012. The methylation level at a given site or region is calculated by dividing the number of reads supporting methylated cytosine (mC) by the total number of reads covering that site or region (which includes both methylated and unmethylated cytosines). This is represented by the formula: $\text{Methylation Level (ML)} = \text{mC} / (\text{mC} + \text{umC})$. For example, if a particular site has 10 covering reads, with 2 reads supporting methylation and 8 reads not supporting methylation, the methylation level for that site would be calculated as $2 / (2 + 8) = 0.2$, indicating a 20% methylation level at that specific site.

Q4: What is the process of mC calling, and why is it performed?

A4: mC calling refers to the identification of methylated cytosine sites. In most cases, bisulfite treatment achieves a conversion efficiency of approximately 99%. However, not all unmethylated cytosine (C) sites undergo complete conversion. Determining whether a given site is a true methylated cytosine (mC) follows a binomial distribution. Therefore, we employ a binomial test, $B(n, p)$, to verify the authenticity of methylation at each site.

Here, x represents the number of methylated C reads at a specific site, n denotes the read coverage depth at that site, and p corresponds to the bisulfite non-conversion rate (the probability of an unmethylated C remaining unconverted). The test evaluates whether observing x methylated C reads is statistically significant under the given sequencing depth (n) and non-conversion rate (p). The criteria for methylation site identification are: $q\text{-value} < 0.01$ and $\text{read coverage} \geq 5$.